
Structural Causal Bandits: Where to Intervene?

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Abstract

We study the problem of identifying the best action in a sequential decision-making setting when the reward distributions of the arms exhibit a non-trivial dependence structure, which is governed by the underlying causal model of the domain where the agent is deployed. In this setting, playing an arm corresponds to intervening on a set of variables and setting them to specific values. In this paper, we show that whenever the underlying causal model is not taken into account during the decision-making process, restricting interventions to all variables simultaneously may exclude every optimal arm, regardless of the number of trials. Exploring all subsets retains an optimal arm but may require substantially more samples. We formally acknowledge this phenomenon and investigate structural properties implied by the underlying causal model, which lead to a graphical characterization of possibly-optimal minimal intervention sets. We leverage this characterization to build a new algorithm that takes as input a causal structure and enumerates all possibly-optimal minimal intervention sets in the graph-only model class, retaining an arm with maximal expected reward. We empirically demonstrate that the new strategy learns an optimal policy and improves sample efficiency in the reported experiments.

1 Introduction

The multi-armed bandit (MAB) problem is one of the prototypical settings studied in the sequential decision-making literature [Lai and Robbins, 1985, Even-Dar et al., 2006, Bubeck and Cesa-Bianchi, 2012]. An agent needs to decide which arm to pull and receives a corresponding reward at each time step while keeping the goal of maximizing its cumulative reward in the long run. The challenge is the inherent trade-off between exploiting known arms versus exploring new reward opportunities [Sutton and Barto, 1998, Szepesvári, 2010]. There is a wide range of assumptions underlying MABs, but in most of the traditional settings, no known relation between the arms’ reward distributions is exploited. This is distinct from independence of fresh rewards across rounds. Many strategies were developed to solve this problem, including classic algorithms such as ϵ -greedy, variants of UCB (Auer et al., 2002, Cappé et al., 2013), and Thompson sampling [Thompson, 1933].

Recently, the existence of some non-trivial dependencies among arms has been acknowledged in the literature and studied under the rubric of *structured bandits*, which include settings such as linear [Dani et al., 2008], combinatorial [Cesa-Bianchi and Lugosi, 2012], unimodal [Combes and Proutiere, 2014], and Lipschitz [Magureanu et al., 2014], just to name a few. For example, a linear (or combinatorial) bandit imposes that an action $x_t \in \mathbb{R}^d$ (or $\{0, 1\}^d$) at a time step t incurs a cost $\ell_t^\top x_t$, where ℓ_t is a loss vector chosen by, e.g., an adversary. In this case, an *index-based* MAB algorithm, oblivious to the structural properties, can be suboptimal.

In another line of investigation, rich environments with complex dependency structures are modeled explicitly through the use of causal graphs, where nodes represent decisions and outcome variables, and direct edges represent direct influence of one variable on another [Pearl, 2000]. Despite the

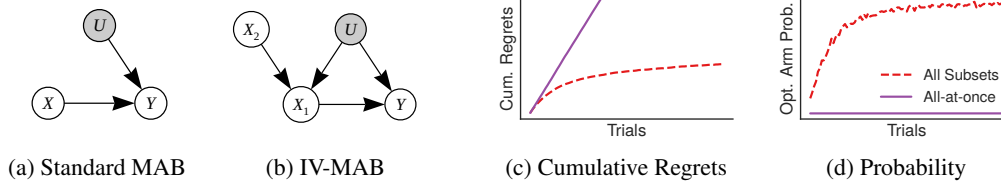


Figure 1: MAB problems as directed acyclic graphs where U is an unobserved variable. Plots of cumulative regrets and probability selecting an optimal arm when a MAB algorithm intervenes X_1 and X_2 simultaneously (All-at-once) or all subsets of $\{X_1, X_2\}$ for IV-MAB. The IV-MAB is also used in the experimental section (see Appendix D [Lee and Bareinboim, 2018] for its parametrization).

apparent connection between MABs and causality, only recently has the use of causal reasoning been incorporated into the design of MAB algorithms. For instance, Bareinboim et al. [2015] first explored the connection between causal models with unobserved confounders (UCs) and reinforcement learning, where latent factors affect both the reward distribution and the player’s intuition. The key observation used in the paper is that while standard MAB algorithms optimize based on the *do*-distribution (formally written as $\mathbb{E}[Y|do(X)]$ or $\mathbb{E}[Y_x]$), the simplest type of counterfactuals, this approach is dominated by another strategy using a more detailed counterfactual as the basis of the optimization process (i.e., $\mathbb{E}[Y_x|X = x']$); this general strategy was called regret decision criterion (RDC). This strategy was later extended to handle counterfactual distributions of higher dimensionality by Forney et al. [2017]. Further, Lattimore et al. [2016] and Sen et al. [2017] studied the problem of best arm identification through importance weighting, where information on how playing arms influences the direct causes (parents, in causal terminology) of a reward variable is available. Zhang and Bareinboim [2017] leveraged causal graphs to solve the problem of off-policy evaluation in the presence of UCs. They noted that whenever UCs are present, traditional off-policy methods can be arbitrarily biased, leading to linear regret. They then showed how to solve the off-policy evaluation problem by incorporating the causal bounds into the decision-making procedure.¹ These works show that ignoring confounding can invalidate observational or off-policy reasoning. Confounding alone does not invalidate stochastic MAB guarantees when the available arms include an optimal intervention and each pull produces a fresh sample from its interventional reward distribution.

In this paper, we focus on the challenge of identifying the best action in MABs where the arms correspond to interventions on an arbitrary causal graph, including when latent variables confound the observed relations (i.e., semi-Markovian causal models). To understand this challenge, we first note that a standard MAB can be seen as the simple causal model as shown in Fig. 1a, where X represents an arm (with K different values), Y the reward variable, and U the unobserved variable that generates the randomness of Y .² After a sufficiently large number of pulls of X (chosen by the specific algorithm), Y ’s average reward can be determined with high confidence.

Whenever a set of UCs affect more than one observed variable, however, novel, non-trivial challenges arise. To witness, consider the more involved MAB structure shown in Fig. 1b, where an unobserved confounder U affects both the action variable X_1 and the reward Y . A naive approach for an algorithm to play such a bandit would be to pull arms in a combinatorial manner, i.e., combining both variables ($X_1 \times X_2$) so that arms are $D(X_1) \times D(X_2)$, where $D(X)$ is the domain of X . One may surmise that this is a valid strategy, albeit not the most efficient one. Somewhat unexpectedly, however, Fig. 1c shows that this is not the case — the optimal action comes from pulling X_2 and ignoring X_1 , while pulling $\{X_1, X_2\}$ together would lead to subpar cumulative rewards (regardless of the number of iterations) since it simply cannot pull the optimal arm (Fig. 1d). After all, if one is oblivious to the causal structure and decides to take all intervenable variables as one (in this case, $X_1 \times X_2$), indiscriminately, one may be doomed to learn a suboptimal policy.

In this paper, we investigate this phenomenon, and more broadly, causal MABs with non-trivial dependency structure between the arms. More specifically, our contributions are as follows: (1) We

¹On another line of investigation, Ortega and Braun [2014] introduced a generalized version of Thompson sampling applied to the problem of adaptive control.

²In causal notation, $Y \leftarrow f_Y(U, X)$, which means that Y ’s value is determined by X and the realization of the latent variable U . If f_Y is linear, we would have a (stochastic) linear bandit. Our results do not constrain the types of structural functions, which is usually within nonparametric causal inference [Pearl, 2000, Ch. 7].

formulate a SCM-MAB problem, which is a structured multi-armed bandit instance within the causal framework. We then derive the structural properties of a SCM-MAB, which are computable from any causal model, including arms' equivalence based on *do*-calculus [Pearl, 1995], and partial orderedness among sets of variables associated with arms in regards to the maximum rewards achievable. (2) We characterize a special set of variables called POMIS (possibly-optimal minimal intervention set), which is worth intervening based on the aforementioned partial orders. We then introduce an algorithm that identifies a complete set of POMISs so that only the subset of arms associated with them can be explored in a MAB algorithm. Simulations corroborate our findings.

Big picture The multi-armed bandit is a rich setting in which a huge number of variants have been studied in the literature. Different aspects of the decision-making process have been analyzed and well-understood in the last decades, which include different functional forms (e.g., linear, Lipschitz, Gaussian process), types of feedback experienced by the agent (bandit, semi-bandit, full), the adversarial or i.i.d. nature of the interactions, just to cite some of the most popular ones. Our study of SCM-MABs puts the causal dimension front and center in the map. In particular, we fully acknowledge the existence of a causal structure among the underlying variables (whenever not known *a priori*, see Footnote 3), and leverage the qualitative relations among them. This is in clear contrast with the prevailing practice that is more quantitative and, almost invariably, is oblivious to the underlying causal structure (as shown in Fig. 1a). We outline in Fig. 2 an initial map that shows the relationship between these dimensions; our goal here is not to be exhaustive, nor prescriptive, but to help to give some perspective. In this paper, we study bandits with no constraints over the underlying functional form (nonparametric, in causality language), i.i.d. stochastic rewards, and with an explicit causal structure acknowledged by the agent.

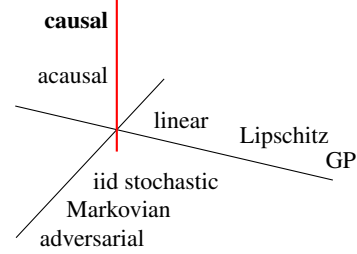


Figure 2: A bandit space with various dimensions (not all dimensions are shown)

Preliminaries: notations and structural causal models

We follow the notation used in the causal inference literature. A capital letter is used for a variable or a mathematical object. The domain of X is denoted by $D(X)$. A bold capital letter is for a set of variables, e.g., $\mathbf{X} = \{X_i\}_{i=1}^n$, while a lowercase letter $x \in D(X)$ is a value assigned to X , and $\mathbf{x} \in D(\mathbf{X}) = \times_{X \in \mathbf{X}} D(X)$. We denote by $\mathbf{x}[\mathbf{W}]$, values of $\mathbf{x}$ corresponding to $\mathbf{W} \cap \mathbf{X}$. A graph $G = \langle \mathbf{V}, \mathbf{E} \rangle$ is a pair of vertices $\mathbf{V}$ and edges $\mathbf{E}$. We adopt family relationships — *pa*, *ch*, *an*, and *de* to denote parents, children, ancestors, and descendants of a given variable; Pa , Ch , An , and De extends *pa*, *ch*, *an*, and *de* by including the argument as the result, e.g., $Pa(X)_G = pa(X)_G \cup \{X\}$. With a set of variables as argument, $pa(\mathbf{X})_G = \bigcup_{X \in \mathbf{X}} pa(X)_G$ and similarly defined for other relations. We denote by $\mathbf{V}(G)$ the set of variables in G . $G[\mathbf{V}']$ for $\mathbf{V}' \subseteq \mathbf{V}(G)$ is a vertex-induced subgraph where all edges among $\mathbf{V}'$ are preserved. We define $G \setminus \mathbf{X}$ as $G[\mathbf{V}(G) \setminus \mathbf{X}]$ for $\mathbf{X} \subseteq \mathbf{V}(G)$.

We adopt the language of Structural Causal Models (SCM) [Pearl, 2000, Ch. 7]. An SCM M is a tuple $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$, where $\mathbf{U}$ is a set of exogenous (unobserved or latent) variables and $\mathbf{V}$ is a set of endogenous (observed) variables. $\mathbf{F}$ is a set of deterministic functions $\mathbf{F} = \{f_i\}$, where f_i determines the value of $V_i \in \mathbf{V}$ based on endogenous variables $\mathbf{PA}_i \subseteq \mathbf{V} \setminus \{V_i\}$ and exogenous variables $\mathbf{U}^i \subseteq \mathbf{U}$, that is, e.g., $v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}^i)$. $P(\mathbf{U})$ is a joint distribution over the exogenous variables. A causal diagram $G = \langle \mathbf{V}, \mathbf{E} \rangle$, associated with M , is a tuple of vertices $\mathbf{V}$ (the endogenous variables) and edges $\mathbf{E}$, where a directed edge $V_i \rightarrow V_j \in \mathbf{E}$ if $V_i \in \mathbf{PA}_j$, and a bidirected edge between V_i and V_j if they share an unobserved confounder, i.e., $\mathbf{U}^i \cap \mathbf{U}^j \neq \emptyset$. Note that $pa(V_i)_G$ corresponds to $\mathbf{PA}_i$. Probability of $Y = y$ when $\mathbf{X}$ is held fixed at $\mathbf{x}$ (i.e., intervened) is denoted by $P(y|do(\mathbf{x}))$, where intervention on $\mathbf{X}$ is graphically represented by $G_{\overline{\mathbf{X}}}$, the graph G with incoming edges onto $\mathbf{X}$ removed. We denote by $CC(X)_G$ the *c-component* of G that contains X where a *c-component* is a maximal set of vertices connected with bidirected edges [Tian and Pearl, 2002]. We define $CC(\mathbf{X})_G = \bigcup_{X \in \mathbf{X}} CC(X)_G$. For a more detailed discussion on the properties of SCMs, we refer readers to [Pearl, 2000, Bareinboim and Pearl, 2016]. Proofs and supplementary material are included in the full version [Lee and Bareinboim, 2018].

Scope of the graphical results. We use acyclic SCMs with independent exogenous sources; any shared source is represented in the graph. Compatibility permits a structural function to ignore a permitted parent and does not impose faithfulness. The existential claims defining graph-only POMISs range over compatible models with freely chosen finite state spaces. In particular, the constructive proofs may enlarge these spaces; they do not establish completeness for prescribed cardinalities, such as an all-binary restriction. Expectations are assumed finite, and maxima are attained. The bandit guarantees below additionally require finitely many arms and bounded rewards.

2 Multi-armed bandits with structural causal models

We recall that MABs consider a sequential decision-making setting where pulling one of the K available arms at each round gives the player a stochastic reward from an unknown distribution associated with the corresponding arm. The goal is to minimize (maximize) the cumulative regret (reward) after T rounds. The mean reward of an arm a is denoted by μ_a and the maximal reward is $\mu^* = \max_{1 \leq a \leq K} \mu_a$. We focus on the cumulative regret, $\text{Reg}_T = T\mu^* - \sum_{t=1}^T \mathbb{E}[Y_{A_t}] = \sum_{a=1}^K \Delta_a \mathbb{E}[T_a(T)]$, where A_t is the arm played at time t , $T_a(t)$ is the number of times arm a has been played after t rounds, and $\Delta_a = \mu^* - \mu_a$.

We now can explicitly connect a MAB instance to its SCM counterpart. Let M be a SCM $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$ and $Y \in \mathbf{V}$ be a reward variable, where $D(Y) \subseteq \mathbb{R}$. The bandit contains arms $\{\mathbf{x} \in D(\mathbf{X}) \mid \mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}\}$, a set of all possible interventions on endogenous variables except the reward variable. Each arm $A_{\mathbf{x}}$ (or simply $\mathbf{x}$) is associated with a reward distribution $P(Y|do(\mathbf{x}))$ where its mean reward $\mu_{\mathbf{x}}$ is $\mathbb{E}[Y|do(\mathbf{x})]$. We call this setting a SCM-MAB, which is fully represented by the pair $\langle M, Y \rangle$. Throughout this paper, we assume that the causal graph G of M is fully accessible to the agent,³ although its parametrization is unknown: that is, an agent facing a SCM-MAB $\langle M, Y \rangle$ plays arms with knowledge of G and Y , but not of $\mathbf{F}$ and $P(\mathbf{U})$. For simplicity, we denote information provided to an agent playing a SCM-MAB by $\llbracket G, Y \rrbracket$. We now investigate some key structural properties that follow from the causal structure G of the SCM-MAB.

Property 1. Equivalence among arms

We start by noting that *do*-calculus [Pearl, 1995] provides rules to evaluate invariances in the interventional space. In particular, we focus here on the Rule 3, which ascertains the condition such that a set of interventions does not have an effect on the outcome variable, i.e., $P(y|do(\mathbf{x}, \mathbf{z}), \mathbf{w}) = P(y|do(\mathbf{x}), \mathbf{w})$. Since arms correspond to interventions (including the *null* intervention) and there is no contextual information, we consider examining $P(y|do(\mathbf{x}, \mathbf{z})) = P(y|do(\mathbf{x}))$ through $Y \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{X}$ in $G_{\overline{\mathbf{X} \cup \mathbf{Z}}}$, which implies $\mu_{\mathbf{x}, \mathbf{z}} = \mu_{\mathbf{x}}$. If valid, this condition implies that it is sufficient to play only one arm among arms in the equivalence class.

Definition 1 (Minimal Intervention Set (MIS)). A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is said to be a *minimal intervention set* relative to $\llbracket G, Y \rrbracket$ if there is no $\mathbf{X}' \subset \mathbf{X}$ such that $\mu_{\mathbf{x}[\mathbf{X}']} = \mu_{\mathbf{x}}$ for every assignment $\mathbf{x} \in D(\mathbf{X})$ and every SCM conforming to G .

For instance, the MISs corresponding to the causal graphs in Fig. 3 are $\{\emptyset, \{X\}, \{Z\}\}$, which do not include $\{X, Z\}$ since $\mu_x = \mu_{x,z}$. The MISs are determined without considering the UCs in a causal graph. The empty set and all singletons in $\text{an}(Y)_G$ are MISs for G with respect to Y . The task of finding the best arm among all possible arms can be reduced to a search within the MISs.

Proposition 1 (Minimality). A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a minimal intervention set for G with respect to Y if and only if $\mathbf{X} \subseteq \text{an}(Y)_{G_{\overline{\mathbf{X}}}}$.

All the MISs given $\llbracket G, Y \rrbracket$ can be determined without explicitly enumerating $2^{\mathbf{V} \setminus \{Y\}}$ while checking the condition in Prop. 1. We provide an efficient recursive algorithm enumerating the complete set of MISs given G and Y (Appendix A), which runs in $O(mn^2)$ where m is the number of MISs.

³In settings where this is not the case, learning G is a separate problem requiring suitable identifiability and sampling assumptions. Observational data need not determine a unique graph [Spirtes et al., 2001]; experimental procedures also require specific assumptions [Kocaoglu et al., 2017].

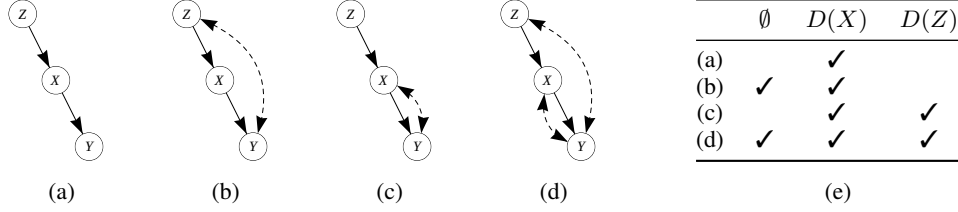


Figure 3: (a–d) Causal graphs such that $\mu_x = \mu_{x,z}$, and (e) non-dominated arms

Property 2. Partial-orders among arms

We now explore the partial-orders among subsets of $\mathbf{V} \setminus \{Y\}$ within the MISs. Given the causal diagram G , it is possible that intervening on some variables is *always* as good as intervening on another set of variables (regardless of the parametrization of the underlying model). Formally, there can be two different sets of variables $\mathbf{W}, \mathbf{Z} \subseteq \mathbf{V} \setminus \{Y\}$ such that

$$\max_{\mathbf{w} \in D(\mathbf{W})} \mu_{\mathbf{w}} \leq \max_{\mathbf{z} \in D(\mathbf{Z})} \mu_{\mathbf{z}}$$

in every possible SCM conforming to G . If that is the case, those arms can be excluded while preserving the optimal reward value. We next define Possibly-Optimal MIS, which incorporates the partial-orderedness among subsets of $\mathbf{V} \setminus \{Y\}$ into MIS denoting the optimal value for a $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ given a SCM by $\mathbf{x}^*$.

Definition 2 (Possibly-Optimal Minimal Intervention Set (POMIS)). Given information $\llbracket G, Y \rrbracket$, let $\mathbf{X}$ be a MIS. If there exists a SCM conforming to G such that $\mu_{\mathbf{x}^*} > \mu_{\mathbf{z}^*}$ for every $\mathbf{Z} \in \mathbb{Z} \setminus \{\mathbf{X}\}$, where $\mathbb{Z}$ is the set of MISs with respect to G and Y , then $\mathbf{X}$ is a *possibly-optimal minimal intervention set* with respect to the information $\llbracket G, Y \rrbracket$.

Intuitively, one may believe that the best action will be to intervene on the direct causes (parents) of the reward variable Y , since this would entail a higher degree of “controllability” of Y within the system. This, in fact, holds true if Y is not confounded with any of its ancestors, which includes the case where no unobserved confounders are present in the system (i.e., Markovian models).

Proposition 2. Given information $\llbracket G, Y \rrbracket$, if Y is not confounded with $an(Y)_G$ via unobserved confounders, then $pa(Y)_G$ is the only POMIS.

Corollary 3 (Markovian POMIS). Given $\llbracket G, Y \rrbracket$, if G is Markovian, then $pa(Y)_G$ is the only POMIS.

For instance, in Fig. 3a, $\{\{X\}\}$ is the set of POMISs. Whenever unobserved confounders (UCs) are present,⁴ on the other hand, the analysis becomes more involved. To witness, let us analyze the maximum achievable rewards of the MISs in the other causal diagrams in Fig. 3. We start with Fig. 3b and note that $\mu_{z^*} \leq \mu_{x^*}$ since $\mu_{z^*} = \sum_x \mu_x P(x|do(z^*)) \leq \sum_x \mu_x P(x|do(z^*)) = \mu_{x^*}$. On the other hand, $\mu_\emptyset$ is not comparable to μ_{x^*} . For a concrete example, consider a SCM where the domains of variables are $\{0, 1\}$. Let U be the UC between Y and Z where $P(U = 1) = 0.5$. Let $f_Z(u) = 1 - u$, $f_X(z) = z$, and $f_Y(x, u) = x \oplus u$, where $\oplus$ is the exclusive-or function. If X is not intervened on, x will be $1 - u$ yielding $y = 1$ for both cases $u = 0$ or $u = 1$ so that $\mu_\emptyset = 1$. However, if X is intervened to either 0 or 1, y will be 1 only half the time since $P(U = 1) = 0.5$, which results in $\mu_{x^*} = 0.5$. We also provide in Appendix A a SCM such that $\mu_\emptyset < \mu_{x^*}$ holds true. This model ($\mu_\emptyset > \mu_{x^*}$) illustrates an interesting phenomenon — allowing an UC to affect Y freely may lead to a higher reward, which may be broken upon interventions. We now consider the different confounding structure shown in Fig. 3c (similar to Fig. 1b), where the variable Z lies outside of the influence of the UC associated with Y . In this case, intervening on Z leads to a higher reward, $\mu_{z^*} \geq \mu_\emptyset$. To witness, note that $\mu_\emptyset = \sum_z \mathbb{E}[Y|z] P(z) = \sum_z \mu_z P(z) \leq \sum_z \mu_{z^*} P(z) = \mu_{z^*}$. However, μ_{z^*} and μ_{x^*} are incomparable, which is shown through two models provided in Appendix A. Finally, we can add the confounders of the two previous models, which is shown in Fig. 3d. In this case, all three μ_{x^*} , μ_{z^*} , and $\mu_\emptyset$ are incomparable. One can imagine scenarios where the influence of the UCs are weak enough so that corresponding models produce results similar to Figs. 3a to 3c.

It’s clear that the interplay between the location of the intervened variable, the outcome variable, and the UCs entails non-trivial interactions and consequences in terms of the reward. The table in

⁴Recall that unobserved confounders are represented in the graph as bidirected dashed edges.

Fig. 3e highlights the arms that are contenders to generate the highest rewards in each model (i.e., each arm intervenes a POMIS to specific values), while a non-POMIS is not required to retain the optimal reward value. Interestingly, the only parent of Y , i.e., X , is not dominated by any other arms in any of the scenarios discussed. In words, this suggests that the intuition that controlling variables closer to Y is not entirely lost even when UCs are present; they are not the only POMIS, but certainly one of them. Given that more complex mechanisms cannot be, in general, ruled out, performing experiments would be required to identify the best arm. Still, the results of the table guarantee that the search can be refined so that MAB solvers can discard structurally redundant or weakly dominated alternatives while retaining an optimal arm. Such a reduction can improve sample efficiency; it does not guarantee smaller regret at every finite horizon.

Interpretation of arm pruning. POMISs preserve the best attainable expected reward. This does not establish an optimal exploration strategy: an excluded intervention can still provide information useful for estimating other arms.

3 Graphical characterization of POMIS

Our goal in this section is to graphically characterize POMISs. We will leverage the discussion in the previous section and note that UCs connected to a reward variable affect the reward distributions in a way that intervening on a variable outside the coverage of such UCs (including no UC) can be optimal — e.g., $\{X\}$ for Fig. 3a, $\emptyset$ for Figs. 3b and 3d, and $\{Z\}$ for Fig. 3c. We introduce two graphical concepts to help characterize this property.

Definition 3 (Unobserved-Confounders’ Territory). Given information $\llbracket G, Y \rrbracket$, let H be $G[An(Y)_G]$. A set of variables $\mathbf{T} \subseteq \mathbf{V}(H)$ containing Y is called an *UC-territory* on G with respect to Y if $De(\mathbf{T})_H = \mathbf{T}$ and $CC(\mathbf{T})_H = \mathbf{T}$.

An UC-territory $\mathbf{T}$ is said to be *minimal* if no $\mathbf{T}' \subset \mathbf{T}$ is an UC-territory. A minimal UC-Territory (MUCT) for G and Y can be constructed by extending a set of variables, starting from $\{Y\}$, alternately updating the set with the c-component and descendants of the set.

Definition 4 (Interventional Border). Let $\mathbf{T}$ be a minimal UC-territory on G with respect to Y . Then, $\mathbf{X} = pa(\mathbf{T})_G \setminus \mathbf{T}$ is called an *interventional border* for G with respect to Y .

The interventional border (IB) encompasses essentially the parents of the MUCT. For concreteness, consider Fig. 4a, and note that $\{W, X, Y, Z\}$ is the MUCT for the causal graph with respect to Y , and the IB is $\{S, T\}$ (marked in pink and blue in the graph, respectively). As its name suggests, MUCT is a set of endogenous variables governed by a set of UCs where, unless the territory is $\{Y\}$, at least one UC is adjacent to the reward variable. Specifically, the reward is determined by values of: (1) the UCs governing the MUCT; (2) a set of unobserved variables (other than the UCs) where each affects an endogenous variable in the MUCT; and (3) the IB. Within the ancestral graph $G[An(Y)_G]$, no UC crosses between the MUCT and its complement so that $\mu_{\mathbf{x}} = \mathbb{E}[Y|\mathbf{x}]$ for IB assignments $\mathbf{x}$ with $P(\mathbf{x}) > 0$. The interventional mean is defined even when this observational conditional is not. We now connect MUCT and IB with POMIS. Let $MUCT(G, Y)$ and $IB(G, Y)$ be, respectively, the MUCT and IB given $\llbracket G, Y \rrbracket$.

Proposition 4. $IB(G, Y)$ is a POMIS given $\llbracket G, Y \rrbracket$.

The main strategy of the proof is to construct a SCM M where intervening on any variable in $MUCT(G, Y)$ causes significant loss of reward. It seems that MUCT and IB can only identify a single POMIS given $\llbracket G, Y \rrbracket$. However, they, in fact, serve as basic units to identify all POMISs.

Proposition 5. Given $\llbracket G, Y \rrbracket$, $IB(G_{\overline{\mathbf{W}}}, Y)$ is a POMIS, for any $\mathbf{W} \subseteq \mathbf{V} \setminus \{Y\}$.

Prop. 5 generalizes Prop. 4 for when $\mathbf{W} \neq \emptyset$ while taking care of UCs across $MUCT(G_{\overline{\mathbf{W}}}, Y)$, and its outside in the original causal graph G . See Fig. 4d, for an instance, where $IB(G_{\overline{W}}, Y) = \{W, T\}$. Intervening on W cuts the influence of S and the UC between W and X , while still allowing the UC to affect X .⁵ Similarly, one can see in Fig. 4b that $IB(G_{\overline{X}}, Y) = \{T, W, X\}$ where intervening on X lets Y be the only element of MUCT making its parents an interventional border,

⁵Note that exogenous variables that do not affect more than one endogenous variable (i.e., non-UCs) are not explicitly represented in the graph.

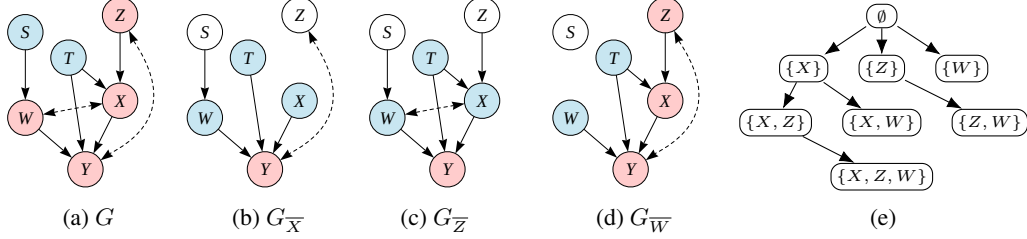


Figure 4: Causal graphs where pink and blue nodes are MUCT and IB, respectively. (Right most) A schematic showing an exploration order of subsets of variables.

Algorithm 1 Algorithm enumerating all POMISs with $\llbracket G, Y \rrbracket$

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1: function POMISs( $G, Y$ )
2:    $\mathbf{T}, \mathbf{X} = \text{MUCT}(G, Y), \text{IB}(G, Y); H = G_{\bar{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ 
3:   return  $\{\mathbf{X}\} \cup \text{subPOMISs}(H, Y, \text{reversed}(\text{topological-sort}(H)) \cap (\mathbf{T} \setminus \{Y\}), \emptyset)$ 
4: function SUBPOMISs( $G, Y, \pi, \mathbf{O}$ )
5:    $\mathbf{P} = \emptyset$ 
6:   for  $\pi_i \in \pi$  do
7:      $\mathbf{T}, \mathbf{X}, \pi', \mathbf{O}' = \text{MUCT}(G_{\pi_i}, Y), \text{IB}(G_{\pi_i}, Y), \pi^{i+1:|\pi|} \cap \mathbf{T}, \mathbf{O} \cup \pi^{1:i-1}$ 
8:     if  $\mathbf{X} \cap \mathbf{O}' = \emptyset$  then
9:        $\mathbf{P} = \mathbf{P} \cup \{\mathbf{X}\} \cup (\text{subPOMISs}(G_{\bar{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}], Y, \pi', \mathbf{O}'))$  if  $\pi' \neq \emptyset$  else  $\emptyset$ 
10:  return  $\mathbf{P}$ 

```

hence, a POMIS. Note that $pa(Y)_G$ is always a POMIS since $\text{MUCT}(G_{\overline{pa(Y)_G}}, Y) = \{Y\}$ and $\text{IB}(G_{\overline{pa(Y)_G}}, Y) = pa(Y)_G$. With Prop. 5, one can enumerate the POMISs given $\llbracket G, Y \rrbracket$ considering all subsets of $\mathbf{V} \setminus \{Y\}$. We show in the sequel that this strategy encompasses all the POMISs.

Theorem 6. *Given $\llbracket G, Y \rrbracket$, $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a POMIS if and only if $\text{IB}(G_{\bar{\mathbf{X}}}, Y) = \mathbf{X}$.*

Thm. 6 provides a graphical necessary and sufficient condition for a set of variables being a POMIS given $\llbracket G, Y \rrbracket$. This characterization allows one to determine all possible arms in a SCM-MAB that are worth intervening on, and, therefore, being free from pulling the other unnecessary arms.

4 Algorithmic characterization of POMIS

Although the graphical characterization provides a means to enumerate the complete set of POMISs given $\llbracket G, Y \rrbracket$, a naively implemented algorithm requires time exponential in $|\mathbf{V}|$. We construct an efficient algorithm (Alg. 1) that enumerates all the POMISs based on Props. 7 and 8 below and the graphical characterization introduced in the previous section (Thm. 6).

Proposition 7. *Let $\mathbf{T}$ and $\mathbf{X}$ be the $\text{MUCT}(G_{\bar{\mathbf{W}}}, Y)$ and $\text{IB}(G_{\bar{\mathbf{W}}}, Y)$, respectively, relative to G and Y . Then, for any $\mathbf{Z} \subseteq \mathbf{V} \setminus \mathbf{T}$, $\text{MUCT}(G_{\bar{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{T}$ and $\text{IB}(G_{\bar{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{X}$.*

Proposition 8. *Let $H = G_{\bar{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ where $\mathbf{T}$ and $\mathbf{X}$ are MUCT and IB given $\llbracket G_{\bar{\mathbf{W}}}, Y \rrbracket$, respectively. Then, for any $\mathbf{W}' \subseteq \mathbf{T} \setminus \{Y\}$, $H_{\bar{\mathbf{W}'}}$ and $G_{\bar{\mathbf{W} \cup \mathbf{W}'}}$ yield the same MUCT and IB with respect to Y .*

Prop. 7 allows one to avoid having to examine $G_{\bar{\mathbf{W}'}}$ for every $\mathbf{W}' \subseteq \mathbf{V} \setminus \{Y\}$. Prop. 8 characterizes the recursive nature of MUCT and IB, where identification of POMISs can be evaluated by subgraphs. Based on these results, we design a recursive algorithm (Alg. 1) to explore subsets of $\mathbf{V} \setminus \{Y\}$ with a certain order. See Fig. 4e for an example where subsets of $\{X, Z, W\}$ are connected based on set inclusion relationship and an order of variables, e.g., (X, Z, W) . That is, there exists a directed edge between two sets if (i) one set is larger than the other by a variable and (ii) the variable's index (as in the order) is larger than other variable's index in the smaller set. The diagram traces how the algorithm will explore the subsets following the edges, while effectively skipping nodes.

Given G and Y , POMISs (Alg. 1) computes a POMIS, i.e., $\text{IB}(G, Y)$. Then, a recursive procedure subPOMISs is called with an order of variables (Line 3). Then subPOMISs examines POMISs by intervening on a single variable against the given graph (Line 6–9). If the IB ($\mathbf{X}$ in Line 7) of such an

Algorithm 2 POMIS-based kl-UCB

```

1: function POMIS-KL-UCB( $B, G, Y, f, T$ )
2:   Input:  $B$ , a SCM-MAB,  $G$ , a causal diagram;  $Y$ , a reward variable
3:    $\mathbf{A} = \bigcup_{\mathbf{X} \in \text{POMISs}(G, Y)} D(\mathbf{X})$ 
4:   kl-UCB( $B, \mathbf{A}, f, T$ )

```

intervened graph intersects with $\mathbf{O}'$ (a set of variables that should be considered in other branch), then no subsequent call is made (Line 8). Otherwise, a subsequent `subPOMISs` call will take as arguments an MUCT-IB induced subgraph (Prop. 8), a refined order, and a set of variables not to be intervened in the given branch. For clarity, we provide a detailed working example in Appendix C with Fig. 4a where the algorithm explores only four intervened graphs ($G, G_{\overline{\{X\}}}, G_{\overline{\{Z\}}}, G_{\overline{\{W\}}}$) and generates the complete set of POMISs $\{\{S, T\}, \{T, W\}, \{T, W, X\}\}$.

Theorem 9 (Soundness and Completeness). *Given information $\llbracket G, Y \rrbracket$, the algorithm `POMISs` (Alg. 1) returns all, and only POMISs.*

The POMISs algorithm can be combined with a MAB algorithm, such as the kl-UCB, creating a simple yet effective SCM-MAB solver (see Alg. 2). For finitely many arms with fresh rewards bounded in $[0, 1]$, and exploration function $f(t) = \log t + 3 \log \log t$ after initialization, kl-UCB satisfies $\limsup_{n \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_n]}{\log(n)} \leq \sum_{\mathbf{x}: \mu_{\mathbf{x}} < \mu^*} \frac{\mu^* - \mu_{\mathbf{x}}}{KL(\mu_{\mathbf{x}}, \mu^*)}$ where KL is Kullback-Leibler divergence between two Bernoulli distributions [Garivier and Cappé, 2011]. Removing suboptimal arms while retaining the optimal value can only decrease this asymptotic upper-bound coefficient. This does not establish finite-horizon regret dominance.

5 Experiments

In this section, we present empirical results demonstrating that the selection of arms based on POMISs makes standard MAB solvers converge faster to an optimal arm. We employ two popular MAB solvers, kl-UCB, which enjoys cumulative regret growing logarithmically with the number of rounds [Cappé et al., 2013], and Thompson sampling (TS, Thompson [1933]), which has strong empirical performance [Kaufmann et al., 2012]. We considered four strategies for selecting arms, including POMISs, MISs, Brute-force, and All-at-once, where Brute-force evaluates all combinations of arms $\bigcup_{\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}} D(\mathbf{X})$, and All-at-once considers intervening in all variables simultaneously, $D(\mathbf{V} \setminus \{Y\})$, oblivious to the causal structure and any knowledge about the action space. The performance of the eight (4×2) algorithms are evaluated relative to three different SCM-MAB instances (the detailed parametrizations are provided in Appendix D). We set the horizon large enough so as to observe near convergence, and repeat each simulation 300 times. We plot (i) the average cumulative regrets (CR) along with their respective standard deviations and (ii) the probability of an optimal arm being selected averaged over the repeated tests (OAP).^{6,7}

Task 1: We start by analyzing a Markovian model. We note that by Cor. 3, searching for the arms within the parent set is sufficient in this case. The number of arms for POMISs, MISs, Brute-force, and All-at-once are 4, 49, 81, and 16, respectively. Note that there are 4 optimal arms within All-at-once arms — for instance, if the parent configuration is $X_1 = x_1, X_2 = x_2$, this strategy will also include combinations of $Z_1 = z_1, Z_2 = z_2, \forall z_1, z_2$. The simulated results are shown in Fig. 5a. CR at round 1000 with kl-UCB are 3.0, 48.0, 72, and 12 (in the order), and all strategies were able to find the optimal arms at this time. POMIS and All-at-once first reached 95% OAP

⁶All the code is available at <https://github.com/sanghack81/SCMMAB-NIPS2018>

⁷One may surmise that combinatorial bandit (CB) algorithms can be used to solve SCM-MAB instances by noting that an intervention can be encoded as a binary vector, where each dimension in the vector corresponds to intervening on a single variable with a specific value. However, the two settings invoke a very different set of assumptions, which makes their solvers somewhat difficult to compare in some reasonably fair way. For instance, the current generation of CB algorithms is oblivious to the underlying causal structure, which makes them resemble very closely the Brute-force strategy, a strategy that can be sample-inefficient for SCM-MABs. Further, the assumption of linearity is arguably one of the most popular considered by CB solvers. The corresponding algorithms, however, will be unable to learn the arms' rewards properly since a SCM-MAB is nonparametric, making no assumption about the underlying structural mechanisms. These are just a few immediate examples of the mismatches between the current generation of algorithms for both causal and combinatorial bandits.

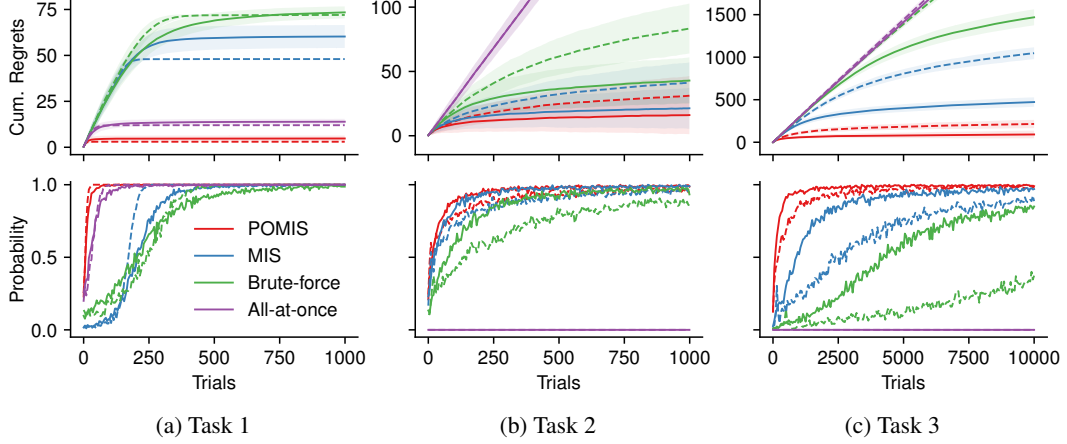


Figure 5: Comparisons across tasks (columns) with cumulative regrets (top) and optimal arm selection probability (bottom) with TS for solid and kl-UCB for dashed lines. Best viewed in color.

at round 20 and 66, respectively. There are two interesting observations at this point. First, at an early stage, OAP for MISs is smaller than Brute-force since it has only 1 optimal arm among 49 arms, while Brute-force has 9 among 81. The advantage of employing MIS over Brute-force is only observed after a sufficiently large number of plays. More interestingly, POMIS and All-at-once both have the common optimal to non-optimal arms-ratio (1:3 versus 4:12), however, POMIS dominates All-at-once since the agent can learn better about the mean reward of the optimal arm while playing non-optimal arms less. Naturally, this translates into less variability and additional certainty about the optimal arm even in Markovian settings.

Task 2: We consider the setting known as instrumental variable (IV), which was shown in Fig. 3c. The optimal arm in this simulation is setting $Z = 0$. The number of arms for the four strategies is 4, 5, 9, and 4, respectively. The results are shown in Fig. 5b. Since the All-at-once strategy only considers non-optimal arms (i.e., pulling Z, X together), it incurs linear regret without selecting an optimal arm (0%). CR (and OAP) at round 1000 with TS are POMIS 16.1 (98.67%), MIS 21.4 (99.00%), Brute-force 42.9 (93.33%), and All-at-once 272.1 (0%). At round 5000, where Brute-force nearly converged, the ratio of CRs for Brute-force and POMIS is $\frac{54.2}{18.1} = 2.99 \gtrsim 2.67 = \frac{9-1}{4-1}$. POMIS, MIS, and Brute-force first reach 95% OAP at 172, 214, and 435.

Task 3: Finally, we study the more involved scenario shown in Fig. 4a. In this case, optimal POMIS arms intervene on $\{S, T\}$, which means that the system should follow its natural flow of UCs, which All-at-once is unable to “pull.” There are 16, 75, 243, and 32 arms for the strategies (in the order). The results are shown in Fig. 5c. The CR (and OAP) at round 10000 with TS are POMIS 91.4 (99.0%), MIS 472.4 (97.0%), Brute-force 1469.0 (85.0%), and All-at-once 2784.8 (0%). The CR ratio at round 10000 is $1469.0/91.4 \approx 16.07$. For the stated parametrization, T cancels between the equations for X and Y , so both $do(S = 0, T = 0)$ and $do(S = 0, T = 1)$ are optimal POMIS arms; $do(S = 0)$ is also optimal among MIS and Brute-force arms. The non-optimal-arm count ratio is therefore $(243 - 3)/(16 - 2) \approx 17.14$. Arm counts alone do not determine the CR ratio, which also depends on reward differences and the solver. Only POMIS and MIS achieved OAP of 95% first in 684 and 3544 steps, respectively.

We start by noticing that the reduction in the CRs is approximately proportional to the reduction in the number of non-optimal arms pulled by (PO)MIS by the corresponding algorithm, which makes the POMIS-based solver the clear winner throughout the simulations. It’s still not inconceivable that the number of arms examined by All-at-once is smaller than for POMIS in a specific SCM-MAB instance, which can favor All-at-once at a finite horizon, though arm counts alone do not determine CR. However, such a lower CR in some instances does not constitute any sort of assurance since arms excluded from All-at-once, but included in POMIS, can be optimal in some SCM-MAB instance conforming to $\llbracket G, Y \rrbracket$. A POMIS-based strategy retains an optimal arm and removes structurally redundant or dominated alternatives. Its finite-horizon regret need not be smaller than that of MIS or Brute-force on every instance. These observations together suggest that, in practice, a POMIS-

based strategy should be preferred when paired with a consistent finite-arm solver under its reward assumptions. The reported experiments show improvements in sample efficiency. Remarkably, there is an interesting trade-off between having knowledge of the causal structure versus not knowing the corresponding dependency structure among arms, and potentially incurring in linear regret (All-at-once) or exponential slow-down (Brute-force). In practice, for the cases in which the causal structure is unknown, the pull of the arms themselves can be used as experiments and could be coupled with efficient strategies to simultaneously learn the causal structure [Kocaoglu et al., 2017].

6 Conclusions

We studied the problem of deciding whether an agent should perform a causal intervention and, if so, which variables it should intervene upon. The problem was formalized using the logic of structural causal models (SCMs) and formalized through a new type of multi-armed bandit called SCM-MABs. We showed that restricting actions to simultaneous interventions on all manipulable variables can exclude every optimal arm. Exhaustive exploration over all intervention subsets avoids that exclusion, but can be sample-inefficient. Learning an unknown causal graph is a separate task requiring additional assumptions. We introduced a novel decision-making strategy based on properties following the *do*-calculus, which allowed the removal of redundant arms, and the partial-orders among the sets of variables existent in the underlying causal system, which led to the understanding of the maximum achievable reward of each interventional set. Leveraging this new strategy based on the possibly-optimal minimal intervention sets (called POMIS), we developed an algorithm that decides whether (and if so, where) interventions should be performed in the underlying system. Finally, we showed by simulations that this causally-sensible strategy performs more efficiently and more robustly than its unstructured counterparts in the reported experiments. We hope that formal machinery and the algorithms developed here can help decision-makers to make more principled and efficient decisions.

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Supplementary Material for “Structural Causal Bandits: Where to Intervene?”

Appendix A Multi-armed bandits with structural causal models

Proposition 1 (Minimality). *A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a minimal intervention set for G with respect to Y if and only if $\mathbf{X} \subseteq \text{an}(Y)_{G_{\overline{\mathbf{X}}}}$.*

Proof. (If) Suppose $\mathbf{X} \subseteq \text{an}(Y)_{G_{\overline{\mathbf{X}}}}$ and fix a proper subset $\mathbf{X}' \subset \mathbf{X}$. Construct a binary SCM in which each variable is the logical OR of its parents and its independent fair private noise; permitted confounders may be ignored. Compare $\text{do}(\mathbf{X} = 0)$ with $\text{do}(\mathbf{X}' = 0)$. Monotonicity implies that the former outcome is no larger for every noise realization. Choose $W \in \mathbf{X} \setminus \mathbf{X}'$. There is a directed path from W to Y containing no other member of $\mathbf{X}$. On the positive-probability event that only W ’s private noise is 1, the two outcomes are respectively 0 and 1. The means therefore differ. This rules out every proper subset as a universally equivalent intervention set.

(Only if) Let $\mathbf{X} \not\subseteq \text{an}(Y)_{G_{\overline{\mathbf{X}}}}$. Let $\mathbf{Z} = \mathbf{X} \setminus \text{an}(Y)_{G_{\overline{\mathbf{X}}}}$, which is a nonempty set and $\mathbf{X}' = \mathbf{X} \setminus \mathbf{Z}$. By Rule 3 of *do*-calculus, $\mu_{\mathbf{X}', \mathbf{Z}} = \mu_{\mathbf{X}'}$, which violates the definition of MIS. $\square$

We present an algorithm (Alg. 3) for enumerating all the MISs given a causal diagram G and a reward variable Y . The algorithm builds a set of MISs by adding a variable to a previously obtained MIS so that the resulting set is a MIS.

Algorithm 3 Minimal Intervention Set Enumeration

```

1: function MISs( $G, Y$ )
2:   Input:  $G$  a causal diagram;  $Y$  an outcome variable
3:    $H = G[\text{An}(Y)_G]$ 
4:   return subMISs( $H, Y, \emptyset$ , reversed(topological-sort( $H$ ))  $\cap (\mathbf{V} \setminus \{Y\})$ )

5: function subMISs( $G, Y, \mathbf{X}, \mathbf{W}$ )
6:    $\mathbb{X} = \{\mathbf{X}\}$ 
7:   for  $W_i \in \mathbf{W}$  do
8:      $H = G_{\overline{W_i}}[\text{An}(Y)_{G_{\overline{W_i}}}]$ 
9:      $\mathbb{X} = \mathbb{X} \cup \text{subMISs}(H, Y, \mathbf{X} \cup \{W_i\}, \mathbf{W}^{i+1} \cap \mathbf{V}(H))$ 
10:  return  $\mathbb{X}$ 

```

We prove universal domination for Prop. 2 below. The $\mathbf{T} = \{Y\}$ construction in the proof of Prop. 4 supplies a single SCM in which $\text{pa}(Y)_G$ strictly outperforms every other MIS, establishing its POMIS membership as well.

Proposition 2. *Given information $\llbracket G, Y \rrbracket$, if Y is not confounded with $\text{an}(Y)_G$ via unobserved confounders, then $\text{pa}(Y)_G$ is the only POMIS.*

Proof. Let $\mathbf{X}$ be a MIS. Let $\mathbf{X}' = \mathbf{X} \setminus \text{pa}(Y)_G$ and $\mathbf{Z} = \text{pa}(Y)_G \setminus \mathbf{X}$.

$$\begin{aligned}
\mu_{\mathbf{X}} &= \sum_{\mathbf{z}} \mathbb{E}[Y \mid \text{do}(\mathbf{x}), \mathbf{z}] P(\mathbf{z} \mid \text{do}(\mathbf{x})) \\
&= \sum_{\mathbf{z}} \mathbb{E}[Y \mid \text{do}(\mathbf{x}, \mathbf{z})] P(\mathbf{z} \mid \text{do}(\mathbf{x})) && \because \text{Rule 2} \\
&= \sum_{\mathbf{z}} \mathbb{E}[Y \mid \text{do}(\mathbf{x}[\text{pa}(Y)_G], \mathbf{z})] P(\mathbf{z} \mid \text{do}(\mathbf{x})) \\
&\leq \sum_{\mathbf{z}} \mu_{\text{pa}_Y^*} P(\mathbf{z} \mid \text{do}(\mathbf{x})) \\
&= \mu_{\text{pa}_Y^*}
\end{aligned}$$

□

We describe two models for Fig. 3b showing $\mu_\emptyset > \mu_{x^*}$ and $\mu_\emptyset < \mu_{x^*}$, respectively. Let $\oplus$ represent the exclusive-or function.

- $\mu_\emptyset > \mu_{x^*}$: Let the domains of U , X , and Z be $\{0, 1\}$ and let $0 < P(U = 1) = \alpha < 1$. $\mathbf{F}$ consists of $f_Z(u) = 1 - u$, $f_X(z) = z$, and $f_Y(x, u) = x \oplus u$. Then, $\mu_{x^*} = \mu_{z^*} = \max(\alpha, 1 - \alpha)$, which is smaller than $\mu_\emptyset = 1$:

$$\begin{aligned}
\mu_{z^*} &= \sum_{u, x, y} y \cdot P(y|x, u) P(x|z^*) P(u) \\
&= \sum_{u, x} P(y = 1|x, u) P(x|z^*) P(u) \\
&= \sum_u P(y = 1|X = z^*, u) P(u) \\
&= \alpha P(y = 1|X = z^*, 1) + (1 - \alpha) P(y = 1|X = z^*, 0) \\
&= \alpha \delta_{z^*, 0} + (1 - \alpha) \delta_{z^*, 1} \\
&= \max(\alpha, 1 - \alpha)
\end{aligned}$$

Since $\mu_\emptyset = 1$, observation is strictly better than intervening on either Z or X .

- $\mu_\emptyset < \mu_{x^*}$: Changing $f_Y(x, u)$ to $x + u$, we observe $\mu_{x^*} = \mu_{z^*} = 1 + \alpha > \mu_\emptyset = 1$.

The strict inequalities persist under sufficiently small perturbations of the structural kernels, because the intervention means in these finite models are continuous in their parameters. This also covers the non-binary outcome $Y = x + u$; an XOR perturbation is not appropriate for that outcome without specifying an encoding.

We devise two models where $\mu_{z^*} > \mu_{x^*}$ and $\mu_{z^*} < \mu_{x^*}$, respectively, for Fig. 3c. Let U_X , U_Y , and U_Z be variable-specific exogenous variables affecting X , Y , and Z , respectively. Let U be the unobserved confounder between X and Y .

- $\mu_{z^*} > \mu_{x^*}$: Let $P(U = 1) = 0.5$ with $\forall U_V \in \{U_X, U_Y, U_Z\} P(U_V = 1) = \epsilon \approx 0$. Let $f_Y(x, u, u_Y) = x \oplus (1 - u) \oplus u_Y$, $f_X(z, u, u_X) = z \oplus u_X \oplus u$, and $f_Z(u_Z) = u_Z$. Then, $\mu_x = (1 - \epsilon)P(U = x) + \epsilon P(U = 1 - x) = 0.5$ while $\mu_{z^*} = (1 - \epsilon)^2 + \epsilon^2 = 1 - 2\epsilon + 2\epsilon^2 \approx 1$.
- $\mu_{z^*} < \mu_{x^*}$: Let probabilities of exogenous variables being 1 be 0.5. Let $f_Y(x, u, u_Y) = x + u + u_Y$, $f_X(z, u, u_X) = z \oplus u_X \oplus u$, and $f_Z(u_Z) = u_Z$. Then, $\mu_x = x + 0.5 + 0.5 = x + 1$ and $\mu_z = P(x = 1|do(z)) + 0.5 + 0.5$. Therefore, $\mu_{z^*} = 1.5 < 2 = \mu_{x^*}$.

Appendix B Graphical characterization of POMIS

We present an algorithm (Alg. 4) retrieving a MUCT given $\llbracket G, Y \rrbracket$.

Algorithm 4 Minimal Unobserved Confounders' Territory

```

1: function MUCT( $G, Y$ )
2:    $H = G[An(Y)_G]$ 
3:    $\mathbf{Q} = \{Y\}$ ;  $\mathbf{T} = \{Y\}$ 
4:   while  $\mathbf{Q} \neq \emptyset$  do
5:     remove an element  $Q_1$  from  $\mathbf{Q}$ 
6:      $\mathbf{W} = CC(Q_1)_H$ ;  $\mathbf{T} = \mathbf{T} \cup \mathbf{W}$ ;  $\mathbf{Q} = (\mathbf{Q} \cup de(\mathbf{W})_H) \setminus \mathbf{T}$ 
7:   return  $\mathbf{T}$ 

```

The following proposition and corollary will be used partly to prove propositions and theorems in the main text.

Proposition 10 (Subsumption). *Let $\mathbf{T}$ and $\mathbf{X}$ be the MUCT and IB for G with respect to Y , respectively. Then, for any $\mathbf{Z} \subseteq \mathbf{V} \setminus \{Y\}$, $\mu_{\mathbf{S}^*} \geq \mu_{\mathbf{Z}^*}$ where $\mathbf{S} = (\mathbf{T} \cap \mathbf{Z}) \cup \mathbf{X}$.*

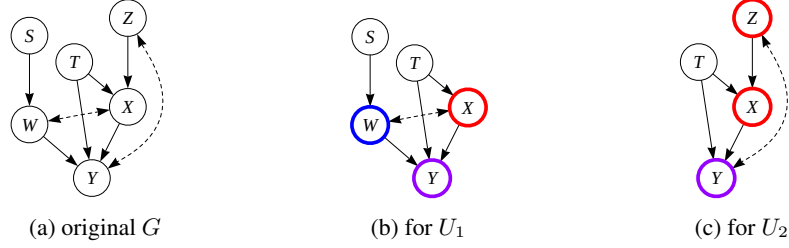


Figure 6: A causal diagram and colored graphs for unobserved confounders.

Proof. (Case $\mathbf{Z} \supseteq \mathbf{X}$) By definition of IB and Rule 3 of *do*-calculus, $\mu_{\mathbf{Z}^*} = \mathbb{E}[Y | do(\mathbf{z}^* [\mathbf{T} \cup \mathbf{X}])]$. Since $\mathbf{Z} \cap (\mathbf{T} \cup \mathbf{X}) = \mathbf{S}$, $\mu_{\mathbf{Z}^*} = \mu_{\mathbf{S}^*}$.

(Otherwise) Let $\mathbf{X}' = \mathbf{X} \setminus \mathbf{Z}$. Then,

$$\mu_{\mathbf{Z}^*} = \sum_{\mathbf{x}'} \mathbb{E}[Y | do(\mathbf{z}^*), \mathbf{x}'] P(\mathbf{x}' | do(\mathbf{z}^*))$$

The first term becomes

$$\begin{aligned} \mathbb{E}[Y | do(\mathbf{z}^*), \mathbf{x}'] &= \mathbb{E}[Y | do(\mathbf{z}^*), do(\mathbf{x}')] && \because \text{Rule 2 } (Y \perp\!\!\!\perp \mathbf{X}' \mid \mathbf{Z})_{G_{\mathbf{Z}\mathbf{X}'}} \\ &= \mathbb{E}[Y | do(\mathbf{z}^* [\mathbf{T} \cup \mathbf{X}]), do(\mathbf{x}')] && \because \text{Rule 3} \\ &\leq \mu_{\mathbf{S}^*} && \because (\mathbf{Z} \cap (\mathbf{T} \cup \mathbf{X})) \cup \mathbf{X}' = \mathbf{S} \end{aligned}$$

Finally, $\mu_{\mathbf{Z}^*} \leq \mu_{\mathbf{S}^*}$ because $\sum_{\mathbf{x}'} \mu_{\mathbf{S}^*} P(\mathbf{x}' | do(\mathbf{z}^*)) = \mu_{\mathbf{S}^*}$. $\square$

Corollary 11. Given $\llbracket G, Y \rrbracket$, no POMIS intersects with an $(\mathbf{X})_G \setminus \mathbf{X}$ where $\mathbf{X} = \text{IB}(G, Y)$.

The proposition says that rewards of arms related to intervening on $\mathbf{Z}$ cannot be better than intervening on $\mathbf{Z}$ and the border $\mathbf{X}$ together. Further, since intervening outside of the territory and the border is ineffective, one can intervene only $\mathbf{Z}$ that are inside the territory altogether with the border.

Proposition 4. $\text{IB}(G, Y)$ is a POMIS given $\llbracket G, Y \rrbracket$.

Proof. Let $\mathbf{X}$ be $\text{IB}(G, Y)$. In this proof, every exogenous variable in $\mathbf{U}$ is binary with its domain being $\{0, 1\}$. An easy case is when $\mathbf{T} = \{Y\}$ where $\mathbf{X}$ is the parents of Y in G . Set $Y = \mathbb{I}\{\mathbf{X} = 0\}$ and let every variable other than Y equal its own independent fair private bit, ignoring other permitted inputs. Then $do(\mathbf{X} = 0)$ has mean 1. A distinct MIS must omit a parent of Y : any extra variable in a superset of all parents would be redundant. An omitted parent remains an independent fair bit, so every distinct MIS has optimal mean at most 1/2. Redundant supersets containing all parents can attain the same optimum and are not competitors in the POMIS definition.

Now, we consider a general case where $\mathbf{T} \supset \{Y\}$, that is, there exists at least one unobserved confounder between Y and its ancestors. As a first step, we construct a SCM M compatible with $H = G[\mathbf{T} \cup \mathbf{X}]$ in which $\mathbf{X}$ is strictly optimal among MISs. Redundant intervention sets are not required to have strictly smaller rewards. To do so, introduce one independent fair bit for every bidirected edge of $H[\mathbf{T}]$, used by that edge's two endpoints. This pairwise model is permitted by the graph-only compatibility class, even when the graph also admits models with a shared source having more than two children. Let $\mathbf{U}' = \{U_j\}_{j=1}^m$ be these edge-specific sources; additional permitted inputs will be ignored. Then, those m individual SCMs $\{M_i\}_{i=1}^m$ will be integrated into a single SCM M in which $do(\mathbf{X} = 0)$ strictly outperforms every other MIS.

We proceed to describe M_i for $U_i \in \mathbf{U}'$. Let $B^{(i)}$ and $R^{(i)}$ be two children of U_i . Let

$$H_i = H \left[De \left(\left\{ B^{(i)}, R^{(i)} \right\} \right)_H \cup \left(\mathbf{X} \cap pa \left(De \left(\left\{ B^{(i)}, R^{(i)} \right\} \right)_H \right) \right) \right]$$

with all bidirected edges removed except U_i . Functions for variables in $H[\mathbf{T}]$ will be described below. We label (i.e., color code) vertices in $De(B^{(i)})_H \setminus De(R^{(i)})_H$ as **blue** and $De(R^{(i)})_H \setminus De(B^{(i)})_H$ as **red**, and $De(B^{(i)})_H \cap De(R^{(i)})_H$ as **purple**. Each of $B^{(i)}$ and $R^{(i)}$ perceives that U_i is a parent

colored as **blue** with value u_i and **red** with value $1 - u_i$, respectively. Those **blue**, **red**, **purple** variables are assigned 3 if any of their parents in $\mathbf{X}$ is not 0. Otherwise, their values are determined as follows. For every **blue** and **red** vertex, its corresponding function returns the common value of its parents of the same color and returns 3 if colored parents' values are not homogeneous. For every **purple** vertex, its corresponding function returns 2 if every **blue**, **red**, and **purple** parent is 0, 1, and 2, respectively, and returns 1 if 1, 0, and 1, respectively. For other cases, the function returns 3. The value 3 is an error marker propagated to Y .

For a variable absent from H_i , assign its i th coordinate the constant 0 and ignore that coordinate wherever it is not an argument of the i th structural equation. We now integrate the m SCMs into one. In M_i , two bits are sufficient to represent every variable. Then, we build a unified SCM where each variable in $\mathbf{T}$ is represented with $2m$ bits where a SCM for U_i will take $2i - 1^{\text{th}}$ and $2i^{\text{th}}$ bit (n.b. the right most digit representing $2^0 = 1$ corresponds to the first bit). We then binarize Y by setting 1 if $2i - 1^{\text{th}}$ and $2i^{\text{th}}$ bits are 01 or 10 for every $1 \leq i \leq m$ and 0 otherwise. Let $P(u_i = 1) = 0.5$ for $U_i \in \mathbf{U}$. This unified SCM M provides a core mechanism to output $Y = 1$ if $\text{do}(\mathbf{X} = 0)$ and $Y = 0$ if $\text{do}(\mathbf{X} \neq 0)$.

We justify the comparison with simultaneous interventions. First work in $\tilde{H} = G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$, and consider a competing MIS $\mathbf{S}$ with $\mathbf{I} = \mathbf{S} \cap \mathbf{T} \neq \emptyset$. Put $\mathbf{K} = \text{An}(\mathbf{I})_{\tilde{H}[\mathbf{T}]}$. Since $Y \notin \mathbf{K}$, the complement $\mathbf{T} \setminus \mathbf{K}$ contains Y and is closed under descendants in $\tilde{H}[\mathbf{T}]$. Minimality of the MUCT implies that a bidirected edge crosses from some $R \in \mathbf{T} \setminus \mathbf{K}$ to some $B \in \mathbf{K}$; otherwise that complement would itself be a smaller territory. Let U_i be its fair shared bit. The vertex R has a path to Y avoiding $\mathbf{I}$, whereas B has a descendant $Z \in \mathbf{I}$. By minimality of $\mathbf{S}$, Z also has a path to Y containing no other intervention vertex.

For the i th coordinate, a valid value at Y is either 1 or 2 and determines a bit θ . The associated requirement is blue value θ , red value $1 - \theta$, and purple value 2 when $\theta = 0$ or 1 when $\theta = 1$. This requirement propagates backwards along a directed path whose interior vertices are not intervened on. The two preceding paths have that property, so the requirement reaches both starting vertices. The natural input at R forces $\theta = U_i$, while the fixed, colored coordinate at Z either is invalid or fixes θ to one constant. Thus $Y = 1$ is possible for at most one value of the fair bit U_i , and its probability is at most $1/2$. This argument allows arbitrary simultaneous interventions and does not assume that an error marker cannot be overwritten.

For competitors disjoint from $\mathbf{T}$, give each border variable its own independent fair private bit. A distinct MIS in $\tilde{H}$ omits a border variable. Every border variable has a child in the territory that is colored in at least one component; its nonzero value therefore produces an error reaching Y when the territory is not intervened on. Hence such a competitor also has mean at most $1/2$, while $\text{do}(\mathbf{X} = 0)$ has mean 1.

Finally extend to G by setting every variable outside $\mathbf{T}$ to its own independent fair private bit and ignoring all permitted inputs not used by the core equations. This is allowed by graph compatibility. Any intervention in G has the same outcome distribution as its restriction to $\tilde{H}$, followed by removal of graphically redundant interventions there. If a MIS of G different from $\mathbf{X}$ reduced to $\mathbf{X}$, it would contain $\mathbf{X}$ and extra vertices. No intervened territory vertex can be present in such a reduction: a downstream-most one has a path to Y inside $\mathbf{T}$ avoiding every other intervention and therefore survives removal of redundant interventions. All extra vertices must consequently be outside $\mathbf{T}$. The border intercepts every directed path from those vertices into $\mathbf{T}$, making them redundant already in G , a contradiction. Every competing MIS therefore reduces to a distinct MIS of $\tilde{H}$ and has optimal mean at most $1/2$. $\square$

We provide an example in Fig. 6 illustrating how sub-SCMs are constructed. Further, values of variables for M_1 , M_2 and a unified M are shown in Table 1 with $s = t = 0$.

Proposition 5. *Given $\llbracket G, Y \rrbracket$, $\text{IB}(G_{\overline{\mathbf{W}}}, Y)$ is a POMIS, for any $\mathbf{W} \subseteq \mathbf{V} \setminus \{Y\}$.*

Proof. Let $\mathbf{T} = \text{MUCT}(G_{\overline{\mathbf{W}}}, Y)$ and $\mathbf{X} = \text{IB}(G_{\overline{\mathbf{W}}}, Y)$. No member of $\mathbf{W}$ belongs to $\mathbf{T}$, so the directed and bidirected edges internal to $\mathbf{T}$, and the directed edges into it, are the same as in G . In particular, $\mathbf{X} = \text{pa}(\mathbf{T})_G \setminus \mathbf{T}$.

U_1	U_2	M_1			M_2			M				
		$w^{(1)}$	$x^{(1)}$	$y^{(1)}$	$z^{(2)}$	$x^{(2)}$	$y^{(2)}$	w	z	x	y'	y
0	0	1	0	2	0	0	2	00 01	00 00	00 00	10 10	1
	1				1	1	1		01 00	01 00	01 10	
1	0	0	1	1	0	0	2	00 00	00 00	00 01	10 01	
	1				1	1	1		01 00	01 01	01 01	

Table 1: Values with $s = t = 0$ where values for M are shown as binary. y' represents $4y^{(2)} + y^{(1)}$ before the value is binarized.

Apply the preceding construction to $\tilde{H} = G_{\bar{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$, whose MUCT and border are $\mathbf{T}$ and $\mathbf{X}$. Extend its equations to G exactly as in the final paragraph of that proof: variables outside $\mathbf{T}$ are independent private fair bits, and additional permitted inputs, including dangling confounders, are ignored. The same reduction of competing MISs applies because $\mathbf{X}$ is the full parent border of $\mathbf{T}$ in G . Thus $do(\mathbf{X} = 0)$ has mean 1 and every other MIS has optimal mean at most $1/2$. This proves strict POMIS membership without an undefined binary flip of a multivalued variable. $\square$

Theorem 6. *Given $\llbracket G, Y \rrbracket$, $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a POMIS if and only if $\text{IB}(G_{\bar{\mathbf{X}}}, Y) = \mathbf{X}$.*

Proof. (If part) A special case of Prop. 5 where $\mathbf{W} = \mathbf{X}$.

(Only if part) It suffices to consider a MIS $\mathbf{W} \subseteq \mathbf{V} \setminus \{Y\}$, since every POMIS is a MIS. Let $\mathbf{T}$ and $\mathbf{X}$ be $\text{MUCT}(G_{\bar{\mathbf{W}}}, Y)$ and $\text{IB}(G_{\bar{\mathbf{W}}}, Y)$ and $\mathbf{T}_0$ and $\mathbf{X}_0$ be $\text{MUCT}(G, Y)$ and $\text{IB}(G, Y)$, respectively. We will prove that $\mathbf{W}$ is not a POMIS when $\mathbf{W} \neq \mathbf{X}$. We can limit $\mathbf{W} \subseteq \mathbf{X}_0 \cup \mathbf{T}_0 \setminus \{Y\}$ due to Prop. 10. Let $\mathbf{X}' = \mathbf{X} \setminus \mathbf{W}$. First, we can observe that $\mathbf{W} \subseteq \text{An}(\mathbf{X})_G$ since otherwise $\mathbf{W} \cap \mathbf{T} \neq \emptyset$, which contradicts that $\mathbf{W}$ is neither a descendant of some variable nor confounded in $G_{\bar{\mathbf{W}}}$. Then,

$$\begin{aligned}
\mu_{\mathbf{W}^*} &= \sum_{\mathbf{x}'} \mathbb{E}[Y | do(\mathbf{w}^*), \mathbf{x}'] P(\mathbf{x}' | do(\mathbf{w}^*)) && \because \text{Basic algebra} \\
&= \sum_{\mathbf{x}'} \mathbb{E}[Y | do(\mathbf{w}^*), do(\mathbf{x}')] P(\mathbf{x}' | do(\mathbf{w}^*)) && \because \text{Rule 2 } (Y \perp\!\!\!\perp \mathbf{X}' \mid \mathbf{W})_{G_{\bar{\mathbf{W}}\mathbf{X}'}} \\
&= \sum_{\mathbf{x}'} \mathbb{E}[Y | do(\mathbf{w}^*[\mathbf{X}]), do(\mathbf{x}')] P(\mathbf{x}' | do(\mathbf{w}^*)) && \because \text{Rule 3 } (Y \perp\!\!\!\perp (\mathbf{W} \setminus \mathbf{X}) \mid \mathbf{X})_{G_{\bar{\mathbf{X}}, \mathbf{W} \setminus \mathbf{X}}} \\
&\leq \sum_{\mathbf{x}'} \mu_{\mathbf{X}^*} P(\mathbf{x}' | do(\mathbf{w}^*)) \\
&= \mu_{\mathbf{X}^*}
\end{aligned}$$

Therefore, $\mathbf{W}$ is not a POMIS. $\square$

Appendix C Algorithmic characterization of POMIS

Proposition 7. *Let $\mathbf{T}$ and $\mathbf{X}$ be the $\text{MUCT}(G_{\bar{\mathbf{W}}}, Y)$ and $\text{IB}(G_{\bar{\mathbf{W}}}, Y)$, respectively, relative to G and Y . Then, for any $\mathbf{Z} \subseteq \mathbf{V} \setminus \mathbf{T}$, $\text{MUCT}(G_{\bar{\mathbf{X}} \cup \mathbf{Z}}, Y) = \mathbf{T}$ and $\text{IB}(G_{\bar{\mathbf{X}} \cup \mathbf{Z}}, Y) = \mathbf{X}$.*

Proof. Given that $\mathbf{T}$ is a MUCT for G being intervened on $\mathbf{X}$, additional intervention outside $\mathbf{T}$ does not affect $\mathbf{T}$ being the MUCT and $\mathbf{X}$ being the IB. $\square$

Proposition 8. *Let $H = G_{\bar{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ where $\mathbf{T}$ and $\mathbf{X}$ are MUCT and IB given $\llbracket G_{\bar{\mathbf{W}}}, Y \rrbracket$, respectively. Then, for any $\mathbf{W}' \subseteq \mathbf{T} \setminus \{Y\}$, $H_{\bar{\mathbf{W}'}}$ and $G_{\bar{\mathbf{W}' \cup \mathbf{W}}}$ yield the same MUCT and IB with respect to Y .*

Proof. $G_{\bar{\mathbf{W}' \cup \mathbf{W}}}$ and $H_{\bar{\mathbf{W}'}}$ share the same edges among $\mathbf{T} \cup \mathbf{X}$ except the fact that $\mathbf{X}$ has no parent in $H_{\bar{\mathbf{W}'}}$, which is irrelevant to identifying MUCT $\mathbf{T}$ and IB $\mathbf{X}$ in both diagrams. $\square$

Proposition 12. *If $\mathbf{W}$ is a POMIS, for any $\mathbf{W}' \subset \mathbf{W}$, $\mathbf{W} \setminus \mathbf{W}' \subset \text{MUCT}(G_{\bar{\mathbf{W}'}}), Y) \cup \text{IB}(G_{\bar{\mathbf{W}'}}), Y)$.*

Proof. Write $G_S = G_{\bar{S}}$, $T_S = \text{MUCT}(G_S, Y)$, and $B_S = \text{IB}(G_S, Y)$. Since $W' \subset W$, the ancestral graph $G_W[An(Y)_{G_W}]$ is a subgraph of $G_{W'}[An(Y)_{G_{W'}}]$. The MUCT is the least closure of $\{Y\}$ under bidirected connectivity and directed descendants in that ancestral graph. Each closure step in the smaller graph is contained in the corresponding step in the larger graph, so $T_W \subseteq T_{W'}$.

By Thm. 6, $W = B_W$. For any $w \in W \setminus W'$, the definition of the border therefore gives an edge $w \rightarrow t$ in G_W with $t \in T_W$. This edge is also present in $G_{W'}$, and $t \in T_{W'}$. Thus either $w \in T_{W'}$ or $w \in pa(T_{W'})_{G_{W'}} \setminus T_{W'} = B_{W'}$, proving the claimed inclusion. Since $Y \in T_{W'} \setminus W$, the inclusion is proper. $\square$

We illustrate how the algorithm works with a causal graph G in Fig. 4a. The graph and its manipulated graphs are shown in Fig. 4. Given G and Y , POMISS obtains a MUCT-and-IB induced subgraph with the IB intervened, $G_{\overline{\{S,T\}}}[\{W, X, Y, Z\} \cup \{S, T\}]$ (Line 2), which is the same as G in this example. While recording the first POMIS, the IB for $G_{\emptyset}$, POMISS calls subPOMISS (Line 3) with $(Y, X, Z, T, W, S) \cap \{W, X, Z\} = (X, Z, W)$ for the parameter π assuming (Y, X, Z, T, W, S) is acquired among many reversed topological orders of H (Line 3). This corresponds to requesting subPOMISS to compute the IBs for the passed causal graph with each variable in (X, Z, W) intervened (Lines 6–10). With $G_{\bar{X}}$, its corresponding MUCT and IB are $\{Y\}$ and $\{T, W, X\}$ (Fig. 4b). The IB will be recorded as a POMIS (Line 9) but no subsequent call for subPOMISS will be made since $(X, Z, W)^{i+1} \cap \{Y\} = \emptyset$. Given $G_{\bar{Z}}$, the same set of MUCT and IB is obtained as $G_{\bar{X}}$ (Fig. 4c). It is unnecessary to record the IB $\{T, W, X\}$ since it contains X , which precedes Z in the given order (Line 7, 8). The MUCT and IB given $G_{\bar{W}}$ are $\{X, Y, Z\}$ and $\{T, W\}$ (Fig. 4d), respectively. It will record the IB $\{T, W\}$, which is disjoint to $\{X, Z\}$. With $\{X, Y, Z\}$ as MUCT, the algorithm checks whether subsequent calls are necessary. However, since both X and Z precede W in the order, no recursive call is made. After all, the complete set of POMISS $\{\{S, T\}, \{T, W\}, \{T, W, X\}\}$ given $\llbracket G, Y \rrbracket$ will be returned only examining 4 IBs compared to $2^5=32$. If an order such that W precedes Z is considered, then a recursive call for subPOMISS will be made for $G_{\overline{\{W,Z\}}}$ after examining $G_{\bar{W}}$. It is unclear at this moment how different ordering affects the number of recursive calls.

Theorem 9 (Soundness and Completeness). *Given information $\llbracket G, Y \rrbracket$, the algorithm POMISS (Alg. 1) returns all, and only POMISSs.*

Proof. Fix the reverse-topological order used by the algorithm. All candidate lists are restrictions of that same order. For a set of selected pivots S , write $T_S = \text{MUCT}(G_{\bar{S}}, Y)$ and $B_S = \text{IB}(G_{\bar{S}}, Y)$. Here S contains only the pivots chosen along a recursion path, not the border interventions used to compress its graph. By Prop. 8, each compressed graph represents the same territory and border, including after further interventions within the territory. Every recorded border is therefore a POMIS of the original graph by Prop. 5. This proves soundness.

Two closure facts will be useful. Territories can only shrink when the intervention set grows: both the ancestral graph and the directed and bidirected connections generating the territory lose edges. Additional interventions outside a current territory preserve its territory and border. Indeed, every territory vertex has a directed path to Y inside the territory, and all connections generating its closure are internal; these survive outside interventions, as do all directed edges entering the territory. No new closure connections can be created by edge deletion.

For completeness, fix a target POMIS W , so $\text{IB}(G_{\bar{W}}, Y) = W$ by Thm. 6. We construct a recursion path with the following invariants: $S \subseteq W$; the candidate list contains every member of $W \cap T_S$; the excluded set O is disjoint from W ; and every member of O precedes every current candidate. They hold at the initial call. If $W \cap T_S = \emptyset$, the outside-intervention fact gives

$$B_S = \text{IB}(G_{\bar{W}}, Y) = W,$$

and this border has already been recorded. Otherwise let Q be the earliest member of $W \cap T_S$ in the candidate list. Put $S' = S \cup \{Q\}$ and let O' be O together with the earlier candidates, exactly as in the pseudocode. Then $W \cap O' = \emptyset$.

We show that the branch is not pruned. Suppose $R \in B_{S'} \cap O'$. Since R is a border vertex, it has a directed path $R \rightarrow T \rightarrow \dots \rightarrow Y$ whose vertices after R lie in $T_{S'}$. Every remaining target vertex in that territory follows Q , and hence R , in reverse-topological order. It therefore cannot occur on the path, whose vertices after R precede R . Also $R \notin W$ and target vertices outside the territory do not

occur on this path. Consequently the path survives $do(\mathbf{W})$. But $\mathbf{T}_{\mathbf{W}} \subseteq \mathbf{T}_{\mathbf{S}'}$, so R is outside $\mathbf{T}_{\mathbf{W}}$. The path must enter that territory through its border, which is $\mathbf{W}$, contradicting that the path contains no target vertex. Thus $\mathbf{B}_{\mathbf{S}'} \cap \mathbf{O}' = \emptyset$.

The suffix and territory restrictions preserve all the invariants in the next call. Each selected pivot leaves the territory, so the path terminates and records $\mathbf{W}$. This proves completeness. In particular, at the first call the order filter compares Q only with earlier territory candidates, not with every member of the new border. At later calls the precise filter is $\mathbf{B}_{\mathbf{S}'} \cap \mathbf{O}' = \emptyset$. $\square$

Appendix D Experiments

Task 1: $P(U_{X_1} = 1) = 0.54$, $P(U_{X_2} = 1) = 0.67$, $P(U_Y = 1) = 0.58$, $P(U_{Z_1} = 1) = 0.54$, and $P(U_{Z_2} = 1) = 0.44$, and functions:

$$\begin{aligned} f_{Z_1}(u_{Z_1}) &= u_{Z_1} \\ f_{Z_2}(u_{Z_2}) &= u_{Z_2} \\ f_{X_1}(z_1, z_2, u_{X_1}) &= z_1 \oplus z_2 \oplus u_{X_1} \\ f_{X_2}(z_1, z_2, u_{X_2}) &= 1 \oplus z_1 \oplus z_2 \oplus u_{X_2} \\ f_Y(x_1, x_2, u_Y) &= (x_1 \wedge x_2) \vee u_Y \end{aligned}$$

Task 2: $P(U_X = 1) = 0.11$, $P(U_Y = 1) = 0.15$, $P(U_Z = 1) = 0.6$, and $P(U_{XY} = 1) = 0.51$ and functions:

$$\begin{aligned} f_Z(u_Z) &= u_Z \\ f_X(z, u_X, u_{XY}) &= u_X \oplus u_{XY} \oplus z \\ f_Y(x, u_Y, u_{XY}) &= 1 \oplus u_Y \oplus u_{XY} \oplus x \end{aligned}$$

Task 3: $P(U_S = 1) = 0.45$, $P(U_T = 1) = 0.81$, $P(U_W = 1) = 0.07$, $P(U_X = 1) = 0.06$, $P(U_Y = 1) = 0.06$, $P(U_Z = 1) = 0.05$, $P(U_{WX} = 1) = 0.51$, and $P(U_{YZ} = 1) = 0.54$, and functions:

$$\begin{aligned} f_S(u_S) &= u_S \\ f_T(u_T) &= u_T \\ f_W(s, u_W, u_{WX}) &= u_W \oplus u_{WX} \oplus s \\ f_Z(u_Z, u_{YZ}) &= u_Z \oplus u_{YZ} \\ f_X(t, z, u_X, u_{WX}) &= 1 \oplus t \oplus z \oplus u_X \oplus u_{WX} \\ f_Y(t, w, x, u_Y, u_{YZ}) &= t \oplus w \oplus x \oplus u_Y \oplus u_{YZ} \end{aligned}$$

Changes

Unofficial AI-corrected version · 2026-09-19 · Author review pending

Original publication. Sanghack Lee and Elias Bareinboim. *Structural Causal Bandits: Where to Intervene?* Advances in Neural Information Processing Systems 31, pp. 2568–2578, 2018. Original proceedings record.

These proposed corrections were prepared with AI assistance. They are not a publisher-issued correction; AI assistance does not imply author approval or certify correctness.

Location	Before → After	Assumptions / limits
Definitions 1–2 (notation)	The assignment quantifier is implicit in MIS; the universal quantifier is misplaced in the POMIS inequality → make the former explicit and write the latter as “for every other MIS.”	The intended MIS and POMIS criteria are unchanged: POMIS compares the candidate with all other MISs within one SCM.
§1; graphical results	Witness state-space scope is implicit → state explicitly that existential witnesses may use freely chosen finite domains.	Completeness for prescribed or all-binary domains is not established. Acyclicity and represented exogenous dependence are explicit.
Propositions 4–5; Appendix B	Incomplete treatment of simultaneous interventions and exterior MISs → independent edge-specific fair bits and surviving paths establish the simultaneous comparisons.	One witness handles all competitors. Compatibility permits unused inputs and does not require faithfulness. No binary flip of a packed variable is needed.
Proposition 12; Theorem 9; Appendix C	Infer non-minimality from dominance; compare the pivot with the entire new border → use nested territories for Prop. 12 and excluded territory candidates for the recursion proof.	Both statements and the algorithm’s pruning rule are unchanged. A dominated MIS can fail to be a POMIS without failing minimality.
Abstract; §§1, 5–6	All-subsets exploration may remain suboptimal; POMIS always dominates → all-subsets retains an optimum; POMIS preserves optimal value without universal regret dominance.	Convergence needs a consistent finite-arm solver and its reward assumptions. Confounding alone does not invalidate fresh interventional rewards. Discarded arms may inform other arms.
§4; unknown-graph discussion	Unqualified KL-UCB and graph-learning implications → state the cited bound’s conditions and treat graph learning as a separate task.	Finite arms, fresh rewards in $[0, 1]$, initialization, $f(t) = \log t + 3 \log \log t$. Bound coefficients do not order finite-horizon regret. Graph learning needs further assumptions.
§5, Task 3; Appendix D	One optimal arm, ratio $(243 - 1)/(16 - 1) \rightarrow$ three tied optima, two among POMIS arms; ratio $(243 - 3)/(16 - 2)$.	All three means equal 62476/78125. The explanatory count is corrected; counts alone do not determine regret ratios.
Throughout; Appendices A–B	Incorrect arithmetic, function arguments, indices, and dominance inference → correct formulas and distinguish dominance from non-minimality; fix wording, references, and notation.	Conditional means need positive support. Weak dominance can exclude POMIS status without excluding MIS status.

Experimental scope. Monte Carlo experiments were not rerun; original plots are unchanged. Task 3’s optimal-arm count and explanatory ratio were corrected analytically.

Citation. Cite the original publication above. Add a *note* identifying the *Unofficial AI-corrected version dated 2026-09-19* consulted and this URL: https://www.sanghacklee.me/ai-corrected/SCB2018_AI_corrected_2026-09-19.pdf. Keep the original publication year.

AI review sign-off: gpt-6-astra · 2026-09-19